

AI5707 — PATTERN RECOGNITION AND MACHINE LEARNING

ASSIGNMENT – 4

Multivariate Gaussian Density Functions, Covariance
Matrices and Decision Boundaries

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Assignment	Assignment – 4
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1. Objective

To generate two-dimensional Gaussian datasets for isotropic, diagonal and full covariance matrices; estimate sample covariance matrices, eigenvalues and eigenvectors; visualize constant-density curves and principal axes with consistent axis scaling; and study how Gaussian decision boundaries change when class means and covariance matrices vary.

2. Mathematical Background

Multivariate Gaussian density:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

Constant-density curve:

$$(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) = c.$$

The notebook plots Mahalanobis distances $d = 1, 2, 3$ and also reports the corresponding density values.

Eigenvalue relation:

$$\Sigma \mathbf{v} = \lambda \mathbf{v}.$$

At Mahalanobis radius 1, the principal semi-axis lengths are $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$.

Gaussian discriminant:

$$g_i(\mathbf{x}) = -\frac{1}{2} \ln |\Sigma_i| - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \Sigma_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i),$$

assuming equal class priors.

Decision boundary:

$$g_1(\mathbf{x}) = g_2(\mathbf{x}).$$

Equal covariance gives a linear boundary; unequal covariance generally gives a quadratic boundary.

3. Experimental Parameters

Experiment	Mean / Class parameters	Covariance matrix
Q1 — Isotropic	$\mu = [0, 2]^T$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Q2 — Diagonal	$\mu = [0, 2]^T$	$\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$
Q3 — Full	$\mu = [0, 2]^T$	$\begin{bmatrix} 4 & 1.2 \\ 1.2 & 1 \end{bmatrix}$
Q4A — Equal covariance	$\mu_1 = [0, 0]^T; \mu_2 = [3, 2]^T$	$\Sigma_1 = \Sigma_2 = \begin{bmatrix} 1.5 & 0 \\ 0 & 1 \end{bmatrix}$
Q4B — Different covariance	$\mu_1 = [0, 0]^T; \mu_2 = [3, 2]^T$	$\Sigma_1 = \begin{bmatrix} 2.5 & 0.8 \\ 0.8 & 1 \end{bmatrix}; \quad \Sigma_2 = \begin{bmatrix} 0.8 & -0.3 \\ -0.3 & 2.2 \end{bmatrix}$
Q4C — Same mean	$\mu_1 = \mu_2 = [1, 1]^T$	$\Sigma_1 = \begin{bmatrix} 3 & 0 \\ 0 & 0.7 \end{bmatrix}; \quad \Sigma_2 = \begin{bmatrix} 0.7 & 0 \\ 0 & 3 \end{bmatrix}$

Parameter note:

The supplied Assignment 4 material available for this submission does not include the earlier numerical mean values referenced by the assignment. Therefore the notebook uses the explicitly stated working values $\mu_1 = 0$ and $\mu_2 = 2$ for Q1–Q3. These values are isolated in one parameter cell and can be replaced if the instructor requires the exact Assignment 1 means.

4. Implementation Procedure

- Generate Gaussian samples using a fixed random seed for reproducibility.
- Estimate sample covariance matrices using the unbiased sample covariance (ddof = 1).
- Compute eigenvalues/eigenvectors with symmetric eigendecomposition and sort eigenvalues in descending order.
- Plot constant-density contours from the Mahalanobis quadratic form and overlay the principal axes.
- Use explicit axis limits, unit aspect ratio and 1-unit major ticks so geometric relationships are visually faithful.
- For classification, evaluate the two Gaussian discriminants on a 2-D grid and plot the zero contour of $g_1 - g_2$.
- Print the numerical decision-boundary equation to verify the graphical result.

5. Q1 — Isotropic Covariance

The theoretical covariance is the identity matrix. The estimated covariance is close to identity, with small deviations caused by finite random sampling. The theoretical eigenvalues are both 1, so the ideal constant-density contours are circles.

Estimated covariance:

$$\begin{bmatrix} 0.9995 & -0.0024 \\ -0.0024 & 1.0319 \end{bmatrix}$$

Estimated eigenvalues: 1.0321 and 0.9993.

Estimated semi-axis lengths: 1.0159 and 0.9997.

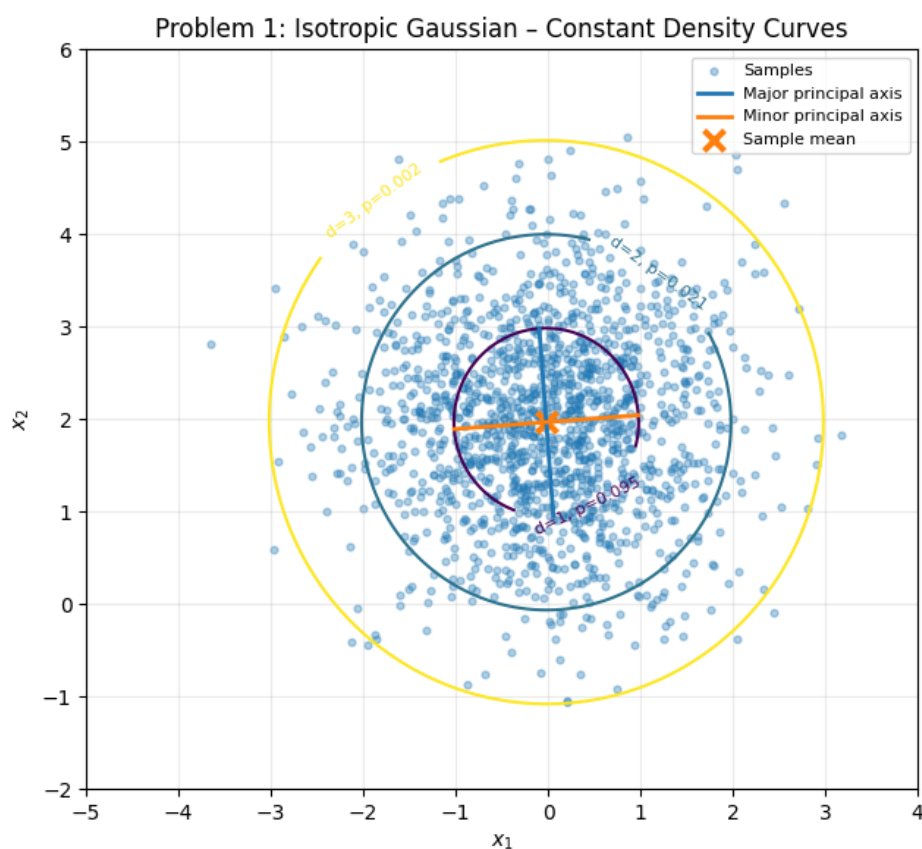


Figure 1: Isotropic Gaussian samples, constant-density contours and principal axes. Contours are labelled by Mahalanobis distance and density.

6. Q2 — Diagonal Covariance

The diagonal covariance has variances 4 and 1 and zero covariance. Therefore the principal axes remain aligned with x_1 and x_2 . The theoretical semi-axis lengths are $\sqrt{4} = 2$ and $\sqrt{1} = 1$, giving a theoretical 2:1 ratio.

Estimated covariance:

$$\begin{bmatrix} 3.7873 & 0.0120 \\ 0.0120 & 1.0474 \end{bmatrix}$$

Estimated eigenvalues: 3.7874 and 1.0474.

Estimated semi-axis lengths: 1.9461 and 1.0234.

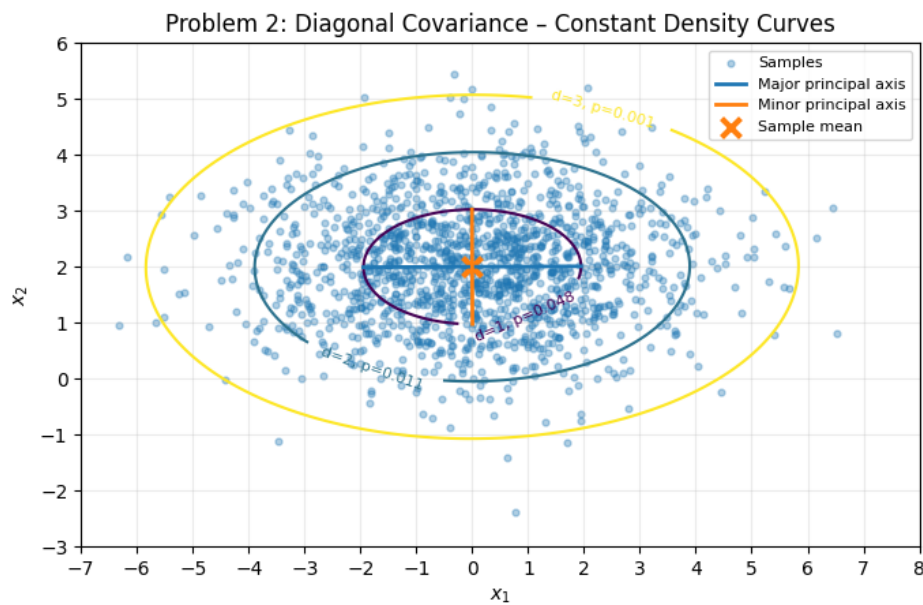


Figure 2: Diagonal covariance: axis-aligned elliptical constant-density contours and principal axes.

7. Q3 — Full Covariance

The non-zero off-diagonal covariance introduces correlation and rotates the principal axes away from the coordinate axes. For the theoretical covariance, the eigenvalues are approximately 4.4209 and 0.5791. The estimated values are 4.3415 and 0.5945, showing close agreement.

Estimated covariance:

$$\begin{bmatrix} 3.9347 & 1.1656 \\ 1.1656 & 1.0013 \end{bmatrix}$$

Estimated semi-axis lengths: 2.0836 and 0.7711.

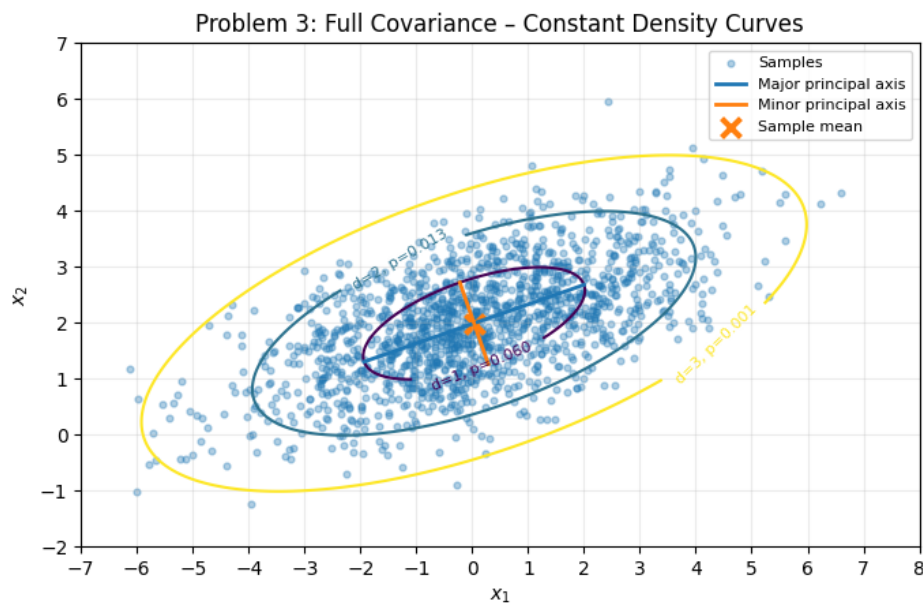


Figure 3: Full covariance: rotated elliptical constant-density contours and principal axes.

8. Theoretical vs Estimated Eigenvalue Comparison

Case	True λ_1, λ_2	Estimated λ_1, λ_2	True axes	Estimated axes	Relative eigenvalue error
Isotropic	1.000, 1.000	1.0321, 0.9993	1.000, 1.000	1.0159, 0.9997	3.21%, 0.07%
Diagonal	4.000, 1.000	3.7874, 1.0474	2.000, 1.000	1.9461, 1.0234	5.32%, 4.74%
Full	4.4209, 0.5791	4.3415, 0.5945	2.1026, 0.7610	2.0836, 0.7711	1.80%, 2.67%

The differences between theoretical and estimated quantities are expected from finite-sample variation. The full-covariance case shows that the eigenvectors capture the rotation while the eigenvalues determine the spread along the rotated axes.

9. Q4 — Gaussian Classes and Decision Boundaries

For each case, the decision boundary is the zero contour of $g_1(\mathbf{x}) - g_2(\mathbf{x})$.

The equal-covariance case produces a straight line, while unequal covariance matrices produce curved boundaries. The same-mean experiment demonstrates that covariance differences alone can create a non-linear boundary.

9.1 Equal covariance — linear boundary

Decision-boundary equation:

$$-2.000x_1 - 2.000x_2 + 5.000 = 0.$$

The quadratic terms cancel because $\Sigma_1 = \Sigma_2$.

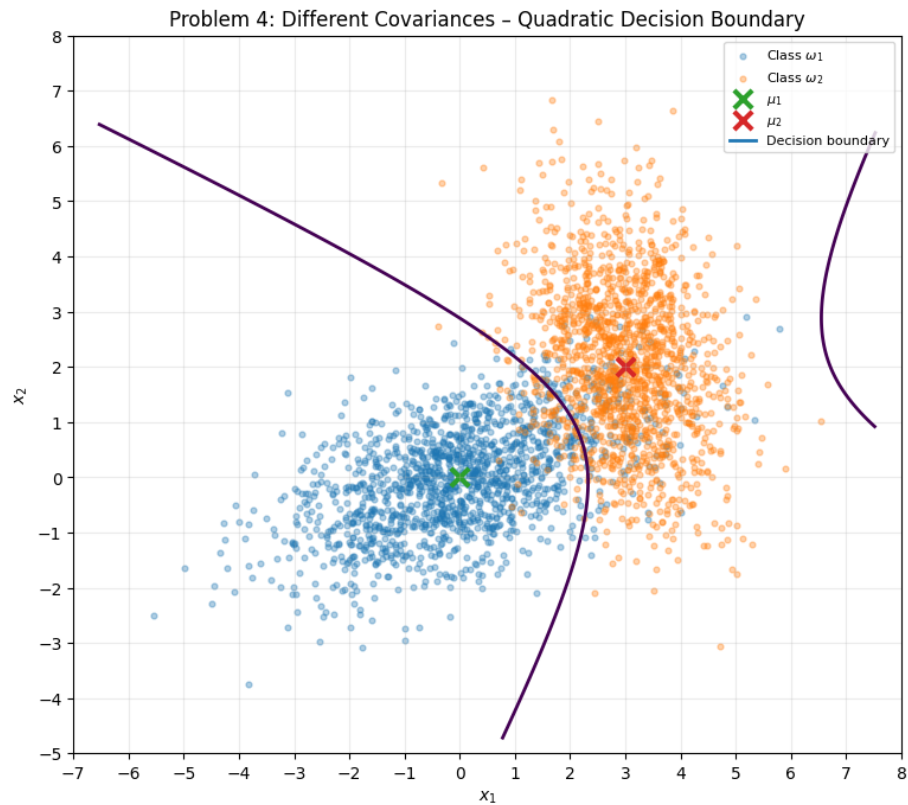


Figure 4: Q1 Equal covariance — linear boundary. Class samples, means and Gaussian decision boundary.

9.2 Different covariance — quadratic boundary

Decision-boundary equation:

$$0.390x_1^2 + 0.610x_1x_2 - 0.433x_2^2 - 4.311x_1 - 1.497x_2 + 7.910 = 0.$$

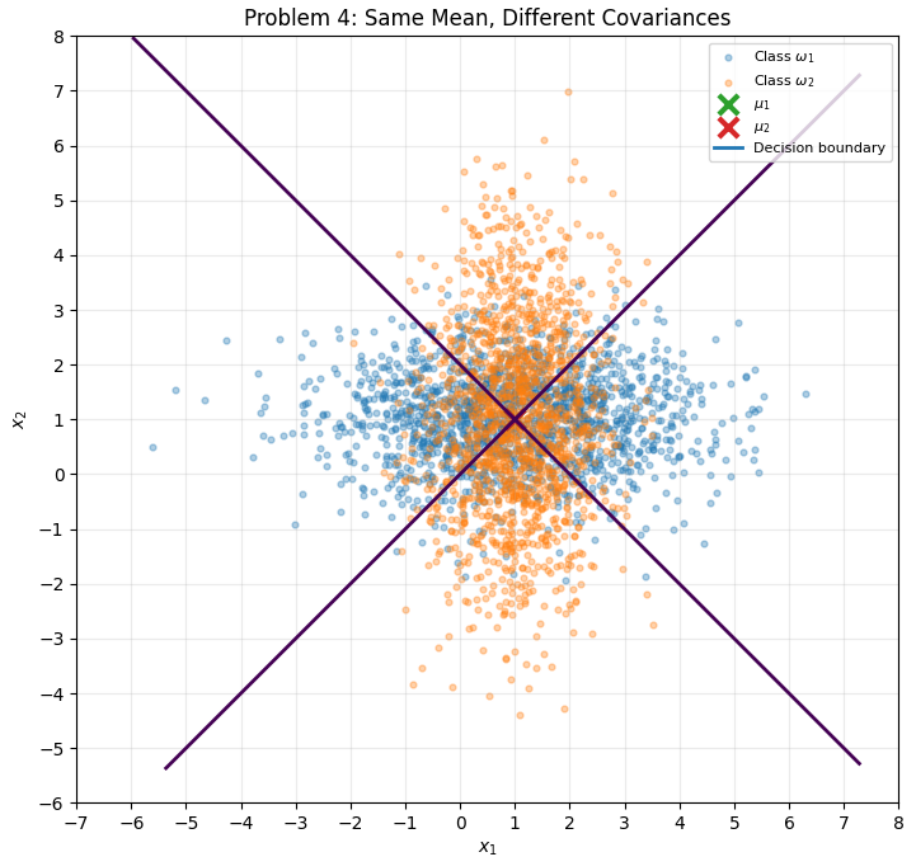


Figure 5: Q2 Different covariance — quadratic boundary. Class samples, means and Gaussian decision boundary.

9.3 Same mean, different covariance

Decision-boundary equation:

$$0.548x_1^2 - 0.548x_2^2 - 1.095x_1 + 1.095x_2 = 0.$$

This confirms that unequal covariance can produce a curved boundary even when $\mu_1 = \mu_2$.

10. Results and Discussion

- **Isotropic covariance:** equal theoretical eigenvalues produce circular density contours. The small deviation from a perfect circle is sampling variation.
- **Diagonal covariance:** zero off-diagonal terms keep the principal axes aligned with the coordinate axes. The larger variance produces the longer axis.
- **Full covariance:** the off-diagonal terms rotate the principal axes. The eigenvectors determine the orientation and the eigenvalues determine the squared spread.
- **Decision boundaries:** equal covariance leads to a linear discriminant boundary; unequal covariance retains quadratic terms and therefore generally produces a curved boundary.
- **Axis scaling:** all figures use equal x/y unit scaling and explicit 1-unit tick spacing, avoiding misleading geometric distortion.
- **Reproducibility:** fixed random seeds are used, so the numerical results and plots can be regenerated consistently.

11. Conclusion

The assignment successfully demonstrates how covariance structure controls the geometry of a multivariate Gaussian distribution. Isotropic covariance produces circular contours, diagonal covariance produces axis-aligned ellipses, and full covariance produces rotated ellipses. Eigenvalues quantify principal spread while eigenvectors identify principal directions. For Gaussian classification, equal covariance matrices lead to a linear decision boundary, whereas unequal covariance matrices generally produce a quadratic boundary. The revised implementation additionally provides consistent axis scaling, density annotations, theoretical-versus-estimated comparisons and explicit numerical boundary equations, making the results easier to verify and interpret.